A Coarse-Graining Approach to Rate Equations of the Composite AC-NBTI Model

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***Abstract*— For ac negative bias temperature instability (NBTI), the composite model that combines the reaction–diffusion (R–D) model for permanent component and the trap–detrap (T–D) model for recoverable component has almost been accepted. However, simple analytical formulas, which are useful for product qualification, have not yet been established. In this paper, we present a coarse-graining approach to analyze the rate equations in the T–D and R–D schemes. Here, the time-averaged rate equations are derived from the equations for stress and recovery phases. The analytical solutions obtained by solving this set of equations can explain the essential features of ac-NBTI, such as the duty-cycle dependence and the memory effect of hole-trapping determined by the standby condition. For the T–D scheme, the exact analytical solutions of the original equations verify this approach, whereas for the R–D scheme, the double-interface R–D model rather than the conventional R–D model captures the duty-cycle dependence observed in the experimental universal curve. Our experimental data on the ac-NBTI are also consistent with the developed analytical model providing a general validity of the approach presented here.**

***Index Terms*— AC negative bias temperature instability (NBTI), coarse-graining, rate equation, reaction–diffusion (R–D), trap–detrap (T–D).**

I. Introduction

NEGATIVE bias temperature instability (NBTI) is one of the most important reliability issues in Si-integrated circuits. To the best of our knowledge, the composite model [1] from the reaction–diffusion (R–D) model for interface-trap formation [2]–[4] and the trap–detrap (T–D) model for hole-trapping [5]–[8] is the most promising one. Though a trapping-only explanation has also been discussed [7], [8], it is not convincing to explain the power-law time dependence and duty-cycle dependence.

However, engineers from other fields will find this model difficult to understand. For example, qualification of products needs the knowledge of the ac effect of NBTI, which might be the key to keep the reliability of the state-of-the-art system-on-chips. Thus, a more simple understanding of the composite model and formulas to estimate the impacts of ac-NBTI are desired.

In addition, the impact of initial conditions of hole-trapping has been neglected in [7]. However, real circuits have the standby mode before the ac operation. Thus, we must consider the memory effect of hole-trapping determined by the standby condition.

The purpose of this paper is to present analytical formulas for the ac-NBTI by analyzing rate equations. First, we derive the rate equations in the T–D scheme for recoverable (R) component in a simple manner such as Shockley–Read–Hall (SRH) statistics [9]. Then, an analytical solution of this equation is discussed. This solution can include the initial condition as well as ac duty cycle and frequency. Moreover, we introduce a coarse-graining approach [10], [11] to the rate equations of the T–D scheme that can explain the essential features of the exact solutions. Second, we extend the same approach to the rate equations of the R–D scheme to obtain the duty-cycle dependence of the permanent (P) component. Since the analytical solution of ac-NBTI is more difficult in the R–D scheme [12], this approach is more important. Finally, we apply these analytical formulas to our ultrafast measure-stress-measure results of ac-NBTI.

II. Rate Equations

A. T–D Model

For the T–D scheme, a capacitor with a time constant is used as a basic unit of the model [5]. However, it is useful to start from the rate equations for T–D processes similar to the R–D scheme. The following rate equations for the areal density of oxide-trapped hole n_{ot} are generally correct for stress and recovery phases. Here, $f_s(t)$ and $f_r(t)$ are the probability of hole occupation in the trap level for the stress and recovery phases, respectively, and D_{ot} is the areal density of the preexisting hole traps [13], [14]. In addition, k_{sf}, k_{sr}, k_{rf}, and k_{rr} are the reaction rate constants

$$\frac{dn_{\mathrm{ot}}}{dt} = k_{sf} D_{\mathrm{ot}}[1 - f_s(t)] - k_{sr} n_{\mathrm{ot}} \tag{1}$$

$$\frac{dn_{\mathrm{ot}}}{dt} = k_{rf} D_{\mathrm{ot}}[1 - f_r(t)] - k_{rr} n_{\mathrm{ot}}. \tag{2}$$

Physical picture of these reactions is shown in Fig. 1, where forward reaction is hole-trapping under $V_{\mathrm{gs}} < 0$ and reverse reaction is hole-detrapping under $V_{\mathrm{gs}} = 0$ [6], [13], [14]. Single-trap level is assumed for simplicity and by considering $n_{\mathrm{ot}}(t) = D_{\mathrm{ot}} f_s(t)$ during the stress and $n_{\mathrm{ot}}(t) = D_{\mathrm{ot}} f_r(t)$

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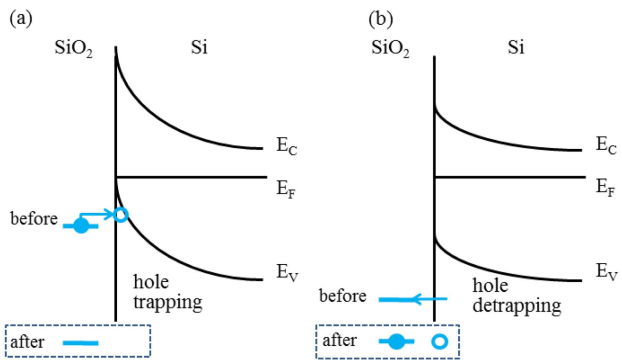

Fig. 1. Schematic of (a) hole-trapping and (b) hole-detrapping process, where the oxide trap having the $\tau_e \ll \tau_c$ nature is shown as an example. The way of illustration is according to [9]. Arrows: direction of electron transition. In the hole-trapping process, some thermal excitation is necessary before the electron-tunneling.

during the recovery, these equations are derived for $f_s(t)$ and $f_r(t)$

$$\frac{df_s(t)}{dt} = k_{sf}[1 - f_s(t)] - k_{sr} f_s(t) \tag{3}$$

$$\frac{df_r(t)}{dt} = k_{rf}[1 - f_r(t)] - k_{rr} f_r(t). \tag{4}$$

In equilibrium

$$0 = k_{sf}(1 - f_s) - k_{sr} f_s \tag{5}$$

$$0 = k_{rf}(1 - f_r) - k_{rr} f_r \tag{6}$$

hold true; hence, we can obtain

$$k_{sr} = \frac{1 - f_s}{f_s} k_{sf} \tag{7}$$

$$k_{rf} = \frac{f_r}{1 - f_r} k_{rr} \tag{8}$$

where f_s and f_r are the Fermi–Dirac-type functions for thermal equilibrium. We assume that (7) and (8) hold in nonequilibrium. This assumption is well known in the SRH statistics [9]. In the cases of high stress, $f_s = 1$ and $f_r = 0$ hold approximately. Considering $k_{sr} = k_{rf} = 0$, we should consider the following equations:

$$\frac{df_s(t)}{dt} = k_{sf}[1 - f_s(t)] \tag{9}$$

$$\frac{df_r(t)}{dt} = -k_{rr} f_r(t). \tag{10}$$

A structural similarity of these equations to those of R and C circuits is the basis of the capacitor model [5]. This type of equation appears in other fields. One example is self-heating phenomena of metal interconnect by ac current.

B. R–D Model

For the R–D scheme, the following rate equations for the interface-trap density N_{it} are well known [2], [3] for the stress and recovery phases, where C_{H} and N_D are the density of released hydrogen at the interface and initial areal density of Si–H bonds, respectively. In addition, A and B denote the reaction rate constants of the following forward and reverse reaction, respectively:

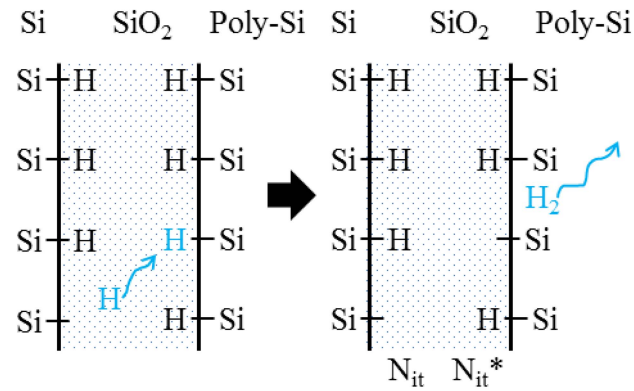

Fig. 2. Schematic of the double-interface R–D model where hydrogen molecules are formed at the SiO_2/Poly-Si interface.

$$\mathrm{Si-H} \Leftrightarrow \mathrm{Si-+H}$$

$$\frac{dN_{\mathrm{it}}}{dt} = A(N_D - N_{\mathrm{it}}) - BN_{\mathrm{it}}C_{\mathrm{H}} \approx AN_D - BN_{\mathrm{it}}C_{\mathrm{H}} \tag{11}$$

$$\frac{dN_{\mathrm{it}}}{dt} = -BN_{\mathrm{it}}C_{\mathrm{H}}. \tag{12}$$

In addition, the equations for the hydrogen atom recombination and hydrogen molecule diffusion are necessary

$$C_{\mathrm{H_2}}(0, t) = kC_{\mathrm{H}}^2 \tag{13}$$

$$\frac{\partial C_{\mathrm{H_2}}}{\partial t} = D\frac{\partial^2 C_{\mathrm{H_2}}}{\partial x^2}. \tag{14}$$

At present, the double-interface R–D model has also attracted attention [1], [15]; thus, we compare the two models.

In the poly-Si/SiO_2 interface, another rate equation is considered [4]

$$\frac{dN_{\mathrm{it}}^*}{dt} = A^* N_D^* C_{\mathrm{H}} - B^* N_{\mathrm{it}}^* C_{\mathrm{H_2}} \tag{15}$$

where N_D^* is the initial areal density of Si–H bonds, and A^* and B^* are the reaction rate constants at the SiO_2/Poly-Si interface. The use of the same C_{H} in (11) and (15) means very fast diffusion of hydrogen atom in SiO_2.

Equation (15) implies the following reaction:

$$\mathrm{Si-H + H} \Leftrightarrow \mathrm{Si-+H_2}$$

as shown in Fig. 2.

In the development of the analytical model of ac-NBTI, successive solutions of (11) and (12) are necessary. Their complicated nature does not allow us to derive simple formulas [12]. Thus, a reduced description is more useful when compared with the T–D model.

III. Devices and Experiments

P-channel MOSFETs with p^+ poly-Si gate were fabricated. The gate dielectrics are 1.6-nm plasma-nitrided SiON. The gate length is 0.1 μm, and its width is 10 μm.

As shown in Fig. 3, the ultrafast measure-stress-measure procedure [15], [16] was used by the Keysight B1500A/B1530A Waveform Generator/Fast Measurement

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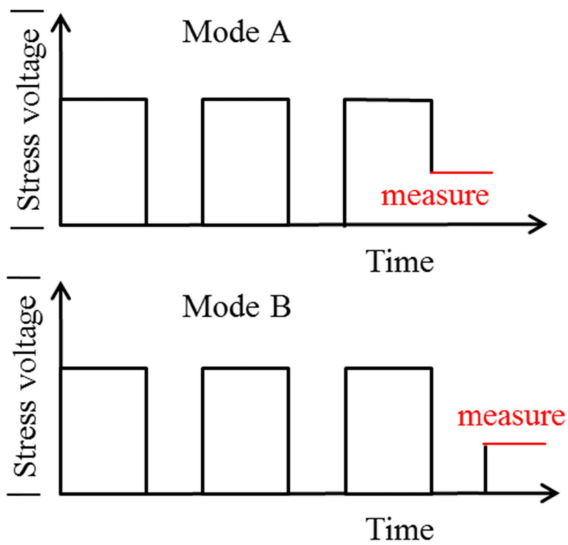

Fig. 3. Schematic of ultrafast measure-stress-measure procedure. Mode A was measured and analyzed in this paper.

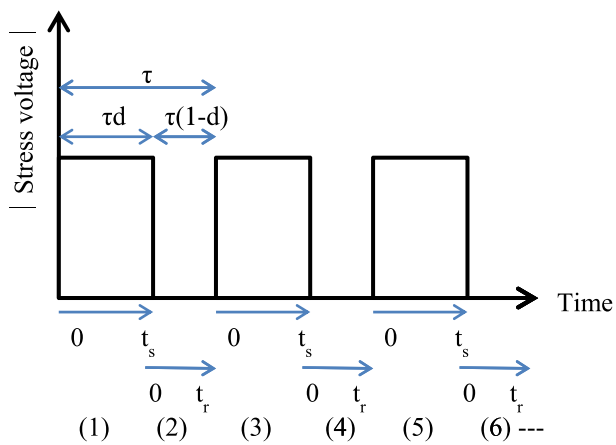

Fig. 4. Pulse profile of stress voltage for the ac-NBTI and time intervals for the analytical calculations.

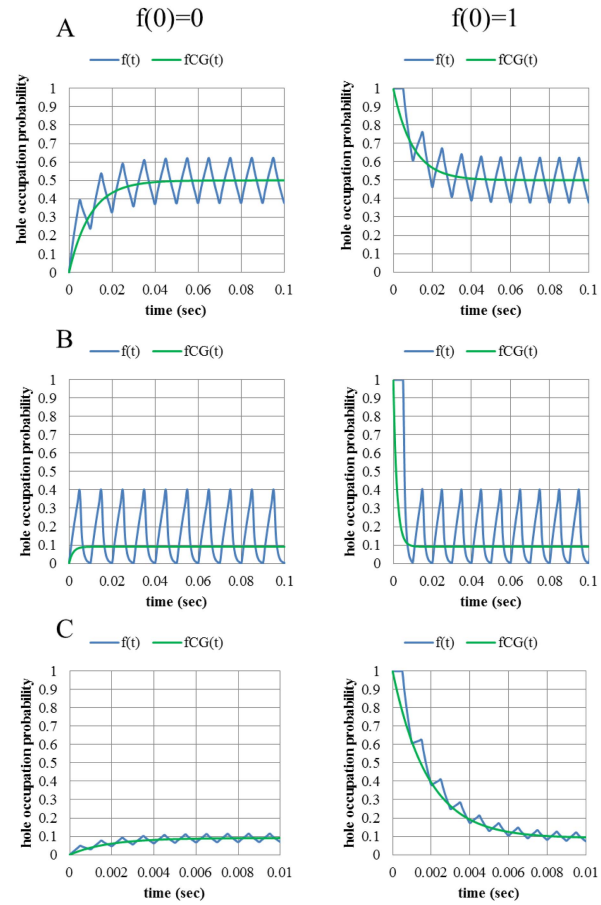

Fig. 5. Oscillatory behavior of hole-occupation probability $f(t)$ in an oxide trap with several parameters shown in Table I under ac-NBT stress on the basis of the T–D model. A(100Hz, τ_e=τ_c), B(100Hz, τ_e<τ_c), C(1kHz, τ_e<τ_c). In these figures, coarse-graining solutions $f_{\mathrm{CG}}(t)$ are also shown, which will be discussed in Section IV-B. The left-hand side graphs correspond to $f(0) = 0$, and the right-hand side graphs correspond to $f(0) = 1$ showing the memory effect of the trapped hole determined by the standby conditions.

Unit (WGFMU) system, in which measurement frequency is 100 Hz and 1 MHz with a measurement delay of $\sim$1 μs. According to [16], Modes A and B shown in Fig. 3 give different results. In this paper, to minimize the recovery, Mode A was measured and analyzed.

IV. Results and Discussion

A. Analytical Solutions of the T–D Model

For the T–D scheme, the equations are simple, and the exact solutions are obtained in the step pulse case, as shown in Fig. 4. In one cycle, we obtain the following solution:

$$f_s(t) = 1 - \Gamma_s(t) + f(0)\Gamma_s(t) \tag{16}$$

$$f_r(t) = f_s(t_s)\Gamma_r(t) \tag{17}$$

where the parameters are defined using pulse frequency f and duty cycle d

$$\tau = \frac{1}{f}, \quad \tau_c = \frac{1}{k_{sf}}, \quad \tau_e = \frac{1}{k_{rr}} \tag{18}$$

$$t_s = \tau d, \quad t_r = \tau(1-d) \tag{19}$$

$$\Gamma_s(t) \equiv e^{-\frac{t}{\tau_c}}, \quad \Gamma_r(t) \equiv e^{-\frac{t}{\tau_e}}. \tag{20}$$

By repeating this calculation for the successive stress and recovery phases shown in Fig. 4, we can obtain analytical solutions (see the Appendix).

In Fig. 5, some solutions of (A5) for different f_0, d, τ_c, and τ_e values shown in Table I are plotted. In the low-frequency cases, oscillatory behavior is observed.

B. Coarse-Graining Approach in the T–D Model

Coarse-graining is an important concept to obtain macroscopic equation from microscopic equations in statistical physics, in which the time interval Δt much larger than τ_C (collision time or correlation time of random force) is introduced [10], [11]. A well-known example is the macroscopic equation of motion for Brownian particle having friction force. This equation can be derived by coarse-graining from the microscopic Newton's equation having time-varying random force [11].

In our case, it is like time-averaging procedure to ignore fast phenomena. In the ac-NBTI, the most important information for reliability is long-term shift of threshold voltage (V_{th}) and

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TABLE I
PARAMETER LISTS OF TIME-DEPENDENT CALCULATION OF THE T–D MODEL (THE DEFINITIONS OF τ_{CG} AND f_{sat} ARE SHOWN IN SECTION IV–B)

	A	B	C
f_0 [Hz]	100	100	1 k
d	0.5	0.5	0.5
τ [s]	1.0E-02	1.0E-02	1.0E-03
τ_c [s]	1.0E-02	1.0E-02	1.0E-02
τ_e [s]	1.0E-02	1.0E-03	1.0E-03
t_r [s]	5.0E-03	5.0E-03	5.0E-04
t_s [s]	5.0E-03	5.0E-03	5.0E-04
τ_{CG} [s]	1.0E-02	1.8E-03	1.8E-03
$f_{sat.}$	0.5	0.091	0.091

pulse duty-cycle dependence for high-frequency circuit operations. For that purpose, a coarse-graining approach is adequate.

In the time interval $\Delta t = N\tau \gg \tau$ ($N \gg 1$ and τ: one period of ac cycle), which is large microscopically but very small macroscopically meaning $\Delta t \ll \tau_c$, τ_e of mainly contributed oxide traps, the change in f is expressed as

$$\begin{aligned}\Delta f &= \Delta f_s + \Delta f_r = [k_{sf}(1-f)d - k_{rr} f(1-d)]\tau N \\ &= \{k_{sf} d - [k_{sf} d + k_{rr}(1-d)] f\}\Delta t \\ &= \left\{\frac{d}{\tau_c} - \left[\frac{d}{\tau_c} + \frac{1-d}{\tau_e}\right] f\right\}\Delta t. \end{aligned} \tag{21}$$

A coarse-graining equation is consequently obtained by averaging Δf by Δt and treating it as coarse-grained time derivative [11]

$$\frac{df_{CG}}{dt} = \frac{d}{\tau_c} - \left[\frac{d}{\tau_c} + \frac{1-d}{\tau_e}\right] f_{CG}. \tag{22}$$

The solution of this equation is

$$f_{CG}(t) = f_{sat}\left(1 - e^{-\frac{t}{\tau_{CG}}}\right) + f(0)e^{-\frac{t}{\tau_{CG}}} \tag{23}$$

$$\frac{1}{\tau_{CG}} \equiv \frac{d}{\tau_c} + \frac{1-d}{\tau_e} \tag{24}$$

$$f_{sat} \equiv f_{CG}(\infty) = \frac{1}{1 + \frac{\tau_c(1-d)}{\tau_e d}}. \tag{25}$$

The essential features of ac-NBTI for high-frequency operation, the duty-cycle dependence of long-term saturated value and the time constant of transient behavior are fully expressed in this formula. Equation (25) can be verified from the exact solution of the T–D model [see (A9)]. As shown in Fig. 5, though $f_{CG}(t)$ is not a simple time average of $f(t)$, it shows gross (macroscopic) behavior of $f(t)s$ in all the cases, in particular Case C.

In Fig. 6, the duty-cycle dependence of f_{sat} is shown. Experimentally observed singular property around high duty [17] can be understood by the fast recovery nature $[(\tau_e/\tau_c) \ll 1]$ of mainly contributing hole traps. This fast recovery nature can be understood in Fig. 1, where hole-trapping from inversion layer needs some thermal excitation before tunneling.

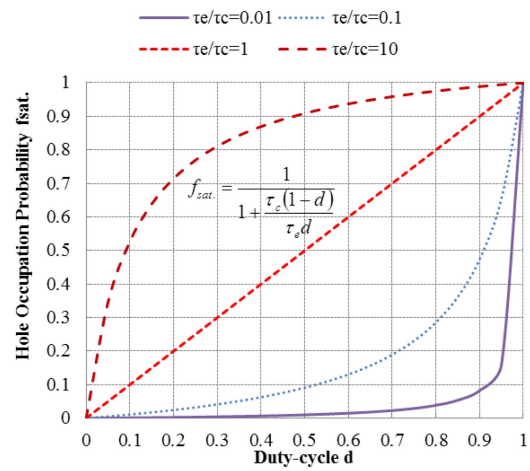

Fig. 6. Duty-cycle dependence of the long-term high-frequency limit values of hole-occupation probability. Experimentally observed singular property around duty = 1 [17] can be understood by the $\tau_e/\tau_c \ll 1$ nature of mainly contributing hole traps.

These hole traps have small effects on a typical 50% ac-NBTI under the initial condition of $f(0) = 0$. However, as shown in Fig. 5, the ac operation of the circuits suffers from the memory effect of trapped holes determined by the standby condition. This impact of the initial condition is clearly expressed in (23). Transient behavior is observed during τ_{CG} which is longer than τ. This memory effect should be considered in designing highly reliable systems. For example, reset pulse ($V_{gs} = 0$) is effective to eliminate this effect.

C. Coarse-Graining Approach in the R–D Model

In the time interval Δt, the change in N_{it} is expressed as

$$\begin{aligned}\Delta N_{it} &= \Delta N_{it,s} + \Delta N_{it,r} \\ &= [(AN_D - BN_{it}C_H)d - BN_{it}C_H(1-d)]\tau N \\ &= (AN_D d - BN_{it}C_H)\Delta t \end{aligned} \tag{26}$$

where Δt is much smaller than the diffusion time scale [4].

A coarse-graining equation is obtained as a result

$$\frac{dN_{it,CG}}{dt} = AN_D d - BN_{it,CG}C_H. \tag{27}$$

According to the well-known procedure [18], a quasi-equilibrium assumption and a simplified triangular solution of diffusion equation for hydrogen molecules are used with (13)

$$AN_D d - BN_{it,CG}C_H \approx 0 \tag{28}$$

$$N_{it,CG} = \frac{1}{2}C_{H_2}\sqrt{Dt} \times 2 = kC_H^2\sqrt{Dt}. \tag{29}$$

As a result, the solution is

$$N_{it,CG}(t) = k^{\frac{1}{3}}\left(\frac{AN_D}{B}d\right)^{\frac{2}{3}}(Dt)^{\frac{1}{6}}. \tag{30}$$

Fig. 7 shows the duty-cycle dependence of N_{it}. Our first coarse-graining solution is not a good approximation to the universal curve on which a lot of the experimental data follows [17].

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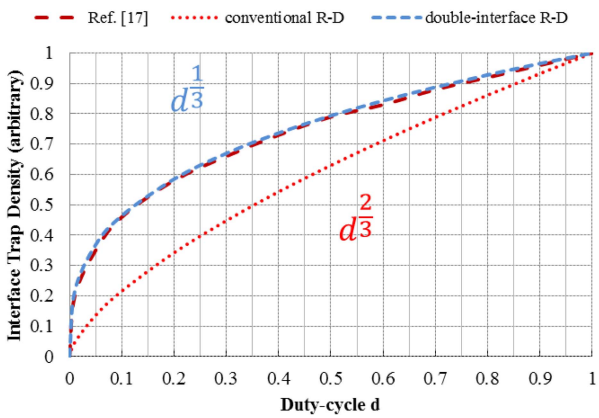

Fig. 7. Duty-cycle dependence of the interface-trap density on the basis of the R–D model. The double-interface R–D model rather than the conventional R–D model captures the duty-cycle dependence reported in [17].

To overcome this discrepancy, the double-interface R–D model [4] is examined. From (15), another quasi-equilibrium condition is obtained

$$A^* N_D^* C_{\mathrm{H}} - B^* N_{\mathrm{it}}^* C_{\mathrm{H}_2} \approx 0. \tag{31}$$

In addition, the number of hydrogen molecules diffused in poly-Si is equal to the total interface-trap density, which is the sum of N_{it} and N_{it}^*

$$N_{\mathrm{it}}^{\mathrm{total}}(t) = C_{\mathrm{H}_2}\sqrt{\mathrm{Dt}}. \tag{32}$$

As a result, a similar equation is obtained from (28), (31), and (32)

$$N_{\mathrm{it}}(t)^{\mathrm{total}} N_{\mathrm{it,CG}}(t) N_{\mathrm{it}}^*(t) = \left(\frac{AA^* N_D N_D^*}{BB^*} d\right)(\mathrm{Dt})^{\frac{1}{2}}. \tag{33}$$

Analyzing the Low-Voltage Stress Induced Leakage Current (LV-SILC) data in [4], we can assume that the time dependence of N_{it} and N_{it}^* is almost the same

$$N_{\mathrm{it}}(t)^{\mathrm{total}} N_{\mathrm{it,CG}}(t) N_{\mathrm{it}}^*(t) \equiv \beta N_{\mathrm{it,CG}}^3(t). \tag{34}$$

As a result

$$N_{\mathrm{it,CG}}(t) = \left(\frac{AA^* N_D N_D^*}{\beta BB^*} d\right)^{\frac{1}{3}} (\mathrm{Dt})^{\frac{1}{6}} \tag{35}$$

is derived. As shown in Fig. 7, the fit to the universal curve has been greatly improved. This duty-cycle dependence $d^{1/3}$ is a very simple and important result of our coarse-graining approach.

D. Comparison With Experiments

In Fig. 8, ultrafast measurement results for the duty-cycle dependence of ac-NBTI are shown together with model-fitting results. The well-known singular property at high duty cycle was also observed in our samples, which means hole trap with the fast recovery nature, as discussed in Section IV-B. This model-fitting to both the 100-Hz and 1-MHz data is done using $f_{\max}(\infty)$ (A6) as the T–D model combined with (35) as the R–D model. Each contribution of the T–D and R–D components is almost 50%, as shown in Fig. 8 (inset).

Though the calculated result using $f_{\mathrm{CG}}(\infty)$ $(=f_{\mathrm{sat}})$ is smaller than that by $f_{\max}(\infty)$ at 100 Hz, both curves are

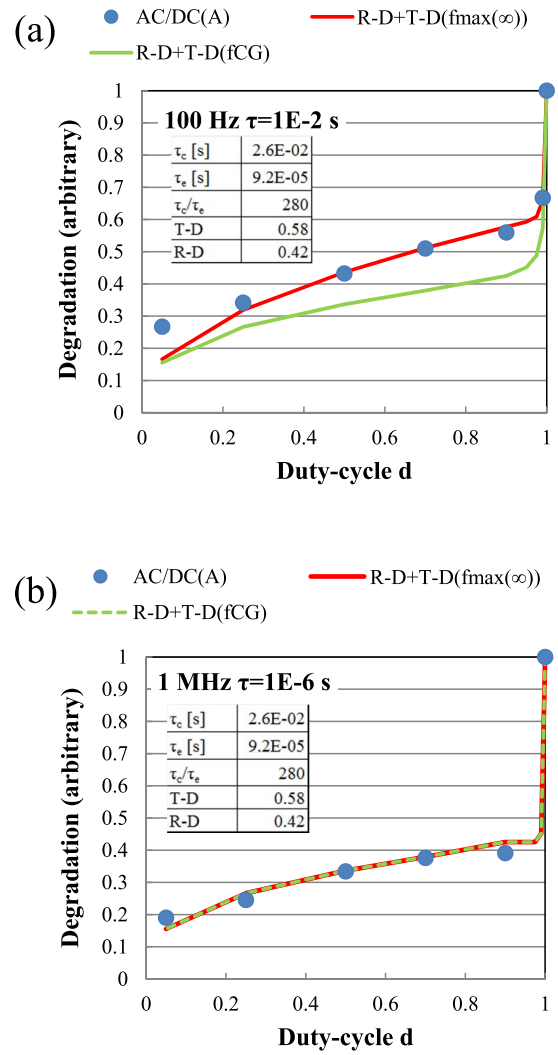

Fig. 8. Duty-cycle dependence of the degradation by the ultrafast measure-stress-measure procedure. (a) 100 Hz. (b) 1-MHz Mode A [16]. Model-fitting on the basis of (A6) combined with (35) is performed. Coarse-graining solution by (25) and (35) can give enough result for 1 MHz.

almost the same at 1 MHz. Hence, it has been confirmed that the f_{CG} solution (23) can give enough result for the high-frequency operation.

It should be noted that the extracted values of τ_e and τ_c are located in the R region of Grasser's Capture/Emission Time (CET) map [8]. Thus, single-trap approximation to the distributed-trap models [5]–[8] is fairly good to explain the duty-cycle dependence, in particular around $d = 1$, as discussed in [6] and [7].

However, the P component is replaced by the R–D model in our composite model. The T–D model for the P component needs extrapolation to very large values of τ_e and τ_c to get the slow recovery nature ($\tau_e > \tau_c$) [7], [8]. Thus, we believe that the R–D model is essential to explain the duty-cycle dependence $d^{1/3}$ of ac-NBTI as well as the long-term behavior $t^{1/6}$ of dc- and ac-NBTI.

V. Conclusion

A coarse-graining approach to rate equations for the ac-NBTI reveals the most important features, such as the duty-

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cycle dependence and memory effect of the initial condition for threshold voltage shift in the high-frequency operation of circuits. For the T–D scheme for R component, the exact solutions of the rate equations verify the above results.

Moreover, for the R–D scheme, the double-interface R–D model can explain the duty-cycle dependence of the P component.

Appendix

In this appendix, the details of analytical solutions for the T–D model are shown. By repeating the calculation for the successive stress and recovery phases shown in Fig. 4, we can obtain analytical solutions

$$\begin{aligned}
(1)\quad & f_1(t) = 1-\Gamma_s(t)+f(0)\Gamma_s(t)\\
(2)\quad & f_2(t) = f_1(t_s)\Gamma_r(t) = [1-\Gamma_s+f(0)\Gamma_s]\Gamma_r(t)\\
(3)\quad & f_3(t) = 1-\Gamma_s(t)+f_2(t_r)\Gamma_s(t)\\
& \quad = 1-\Gamma_s(t)+[(1-\Gamma_s)\Gamma_r+f(0)\Gamma]\Gamma_s(t)\\
(4)\quad & f_4(t) = f_3(t_s)\Gamma_r(t) = [(1-\Gamma_s)(1+\Gamma)+f(0)\Gamma_s\Gamma]\Gamma_r(t)\\
(5)\quad & f_5(t) = 1-\Gamma_s(t)+f_4(t_r)\Gamma_s(t)\\
& \quad = 1-\Gamma_s(t)+[(1-\Gamma_s)\Gamma_r(1+\Gamma)+f(0)\Gamma^2]\Gamma_s(t)\\
(6)\quad & f_6(t) = f_5(t_s)\Gamma_r(t)\\
& \quad = [(1-\Gamma_s)(1+\Gamma+\Gamma^2)+f(0)\Gamma_s\Gamma^2]\Gamma_r(t)\\
(7)\quad & f_7(t) = 1-\Gamma_s(t)+f_6(t_r)\Gamma_s(t)\\
& \quad = 1-\Gamma_s(t)+[(1-\Gamma_s)\Gamma_r(1+\Gamma+\Gamma^2)\\
& \qquad\qquad +f(0)\Gamma^3]\Gamma_s(t)\\
(8)\quad & f_8(t) = f_7(t_s)\Gamma_r(t)\\
& \quad = [(1-\Gamma_s)(1+\Gamma+\Gamma^2)+f(0)\Gamma_s\Gamma^3]\Gamma_r(t)\\
& \quad \ldots
\end{aligned} \tag{A1}$$

where

$$\Gamma_s \equiv \Gamma_s(t_s),\quad \Gamma_r \equiv \Gamma_r(t_r),\quad \Gamma \equiv \Gamma_s\Gamma_r. \tag{A2}$$

In general, we can obtain for $n = 1, 2, 3, \ldots$

$$f_{2n-1}(t) = 1-\Gamma_s(t) + \left[\frac{(1-\Gamma_s)\Gamma_r}{1-\Gamma}(1-\Gamma^{n-1})+f(0)\Gamma^{n-1}\right]\Gamma_s(t) \tag{A3}$$

$$f_{2n}(t) = \left[\frac{1-\Gamma_s}{1-\Gamma}(1-\Gamma^n)+f(0)\Gamma_s\Gamma^{n-1}\right]\Gamma_r(t). \tag{A4}$$

Another expression for total operation time t is more convenient

$$f(t) = \begin{cases} 1-\Gamma_s(t-(n-1)n\tau) \\ +\left[\frac{(1-\Gamma_s)\Gamma_r}{1-\Gamma}(1-\Gamma^{n-1})+f(0)\Gamma^{n-1}\right]\Gamma_s(t-(n-1)\tau) \\ (n-1)\tau < t < (n-1)\tau + t_s \quad (n=1,2,3,\ldots) \\ \left[\frac{1-\Gamma_s}{1-\Gamma}(1-\Gamma^n)+f(0)\Gamma^{n-1}\Gamma_s\right]\Gamma_r(t-(n-1)\tau-t_s) \\ (n-1)\tau + t_s < t < n\tau \quad (n=1,2,3,\ldots) \end{cases} \tag{A5}$$

where $n \equiv$ integer $(t/\tau)+1$.

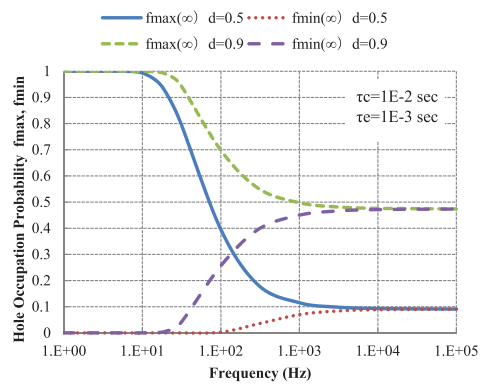

Fig. 9. Frequency dependence of the long-term maximum and minimum values of hole-occupation probability $f_{\max}(\infty)$ and $f_{\max}(\infty)$. At high-frequency limit, both the values approach to $f_{\text{sat}} = f_{\text{CG}}(\infty)$.

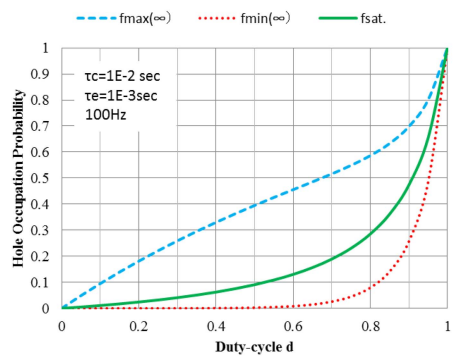

Fig. 10. Duty-cycle dependence of $f_{\max}(\infty)$ and $f_{\max}(\infty)$. In this figure, the values of $f_{\text{sat}} = f_{\text{CG}}(\infty)$ are also shown.

In particular, we are interested in the maximum and minimum values of the oscillations for $t = (n-1)\tau + t_s$ and $t = (n-1)\tau$, respectively, where long-term form and their ripple magnitude are also shown

$$\begin{aligned} f_{\max} &= \frac{1-\Gamma_s}{1-\Gamma}(1-\Gamma^n)+f(0)\Gamma^{n-1}\Gamma_s \\ \rightarrow f_{\max}(\infty) &\equiv \frac{1-\Gamma_s}{1-\Gamma} \end{aligned} \tag{A6}$$

$$\begin{aligned} f_{\min} &= \frac{1-\Gamma_s}{1-\Gamma}\Gamma_r(1-\Gamma^{n-1})+f(0)\Gamma^{n-1} \\ \rightarrow f_{\min}(\infty) &\equiv \frac{1-\Gamma_s}{1-\Gamma}\Gamma_r \end{aligned} \tag{A7}$$

$$\Delta f_{\max-\min}(\infty) = \frac{(1-\Gamma_s)(1-\Gamma_r)}{1-\Gamma}. \tag{A8}$$

In Figs. 9 and 10, the frequency and duty-cycle dependence of long-term $f_{\max}(\infty)$ and $f_{\min}(\infty)$ is shown.

It should be pointed out that at high-frequency limit

$$f_{\max}(\infty), f_{\min}(\infty) \rightarrow f_{\text{sat}} = \frac{1}{1+\frac{\tau_c t_r}{\tau_e t_s}} = \frac{1}{1+\frac{\tau_c(1-d)}{\tau_e d}}. \tag{A9}$$

Moreover, it has been confirmed that $f(t)$ in (A5) is equal to $f_{\text{CG}}(t)$ in (23) in the high-frequency limit [$\tau \rightarrow 0$, $n \rightarrow \infty$, and $t = (n-1)\tau$].

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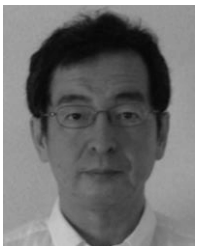

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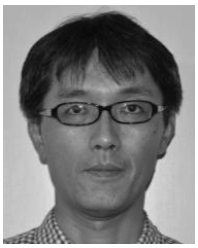

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